Bi-adjoint scalar theory

G Lie groups $\qquad \mathfrak{g}$ Lie algebras

field: $\mathbb{R}^4 \to \mathfrak{g} \otimes \mathfrak{g}$

basis $\{T^a\}$ of $\mathfrak{g}$

structure constants

$$[T^a, T^b] = f^{ab}{}_c T^c$$

$$\Phi(x) = \Phi_{ab}(x)\, T^a \otimes T^b$$

$\to$ matrix

$$\mathcal{L} = \underbrace{\frac{1}{2}\sum_{ab}\left(\partial_\mu \Phi_{ab}\, \partial^\mu \Phi_{ab}\right)}_{\text{kinetic}} + g \sum_{\substack{a_3\\ b_3}} \left(\underbrace{\Phi_{a_1 b_1} \Phi_{a_2 b_2} \Phi_{a_3 b_3}}_{\text{interaction}} f^{a_1 a_2}{}_{a_3} f^{b_1 b_2}{}_{b_3} \right)$$

ex: $\sum_{ab} \phi_{ab}^3$

if a theory has only quadratic terms $\to$ free theory

$$\mathcal{L} = \frac{1}{2}\partial_\mu \phi\, \partial^\mu \phi$$ free scalar field

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in QM: $\quad L = \frac{1}{2} m \sum_i v_i^2 - V(q)$ $\qquad v \in \mathbb{R}^n$, $q \in \mathbb{R}^n$

$q_1^2 + q_2^2 + p_1^2 + p_2^2$

go to Hamiltonian $\to$ $H = \sum_i \frac{p_i^2}{2m} + V(q)$ $\to$ dynamics (classical)

Poisson bracket $\{q_i, p_j\} = \delta_{ij}$

quantize map $q_i \to \hat{q}_i$, $p_i \to \hat{p}_i$ $\qquad$ require $[\hat{q}_i, \hat{p}_j] = i\delta_{ij}$ $\qquad$ representation of Heisenberg group

$+ \hat{H} \to$ dynamics (quantum)

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to extend this to fields think of $\phi(t,x)$ as an infinite (continuous) collection of fields labelled by x

$$\mathcal{L}(q, v) = \frac{1}{2}\eta^{\alpha\beta} v_\alpha v_\beta$$

$q \in \mathbb{R}$, $v \in \mathbb{R}^4$

action $\quad S(\phi) = \int_M d^4x\, \mathcal{L}(\phi(x), \partial\phi(x))$

EoM
$$\boxed{\frac{\partial \mathcal{L}}{\partial q}(\phi(x), \partial\phi(x)) - \frac{\partial}{\partial x^\mu}\frac{\partial \mathcal{L}}{\partial v_\mu}(\phi(x), \partial\phi(x)) = 0}$$

$$0 = \frac{\partial}{\partial x^\mu}\left(\eta^{\mu\nu} v_\nu\right)\Big|_{v=\partial\phi} = \partial_\mu\left(\eta^{\mu\nu}\partial_\nu\phi\right) = \boxed{\partial_\mu\partial^\mu\phi} = \ddot{\phi} - \nabla^2\phi = 0$$

$$\Box\phi = 0$$

$$\frac{\partial\alpha}{\partial s_\mu} = \frac{1}{2}\eta^{\alpha\beta}\frac{\partial}{\partial s_\mu}(s_\alpha s_\beta) = \frac{1}{2}\eta^{\alpha\beta}\left(\frac{\partial s_\alpha}{\partial s_\mu}s_\beta + s_\alpha\frac{\partial s_\beta}{\partial s_\mu}\right)$$

$$= \eta^{\alpha\beta}\delta^\mu_\alpha s_\beta = \eta^{\mu\beta}s_\beta$$

$$S(\phi) = \int dt\; L(\phi(t), \dot{\phi}(t))$$

function function

$$L[q, v] = \frac{1}{2}\int_{\mathbb{R}^3} d^3x\left(v(x)^2 - \nabla q(x)\cdot\nabla q(x)\right)$$

function of $\mathbb{R}^3$

$$v_0^2 - v_1^2 - v_2^2 - v_3^2$$

$$-(\partial_1 q)^2 - (\partial_2 q)^2 - (\partial_3 q)^2$$

$$\phi(t) = \phi(t, \cdot)$$

$$S[\phi] = \int dt\; L(\phi(t), \dot{\phi}(t))$$

$$= \frac{1}{2}\int dt\int d^3x\left(\underbrace{\dot{\phi}(t,x)^2 - \nabla\phi(t,x)\cdot\nabla\phi(t,x)}_{\partial_\mu\phi\,\partial^\mu\phi}\right)$$

δ as a functional.

$$\delta[f] = f(0)$$

<u>Hamiltonian:</u> $\quad p_i = \dfrac{\partial L}{\partial s_i}$

$$F(x) = \sum_i G(x_i)$$

if $\quad F[f] = \displaystyle\int d^3x\; G(f(x))$

How do we define $\dfrac{\delta F}{\delta f}$?

$$dF(\psi)(x)$$

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} F(\underset{\in\mathbb{R}^n}{x} + \varepsilon\underset{\in\mathbb{R}^n}{\psi})$$

$$= \sum_i \frac{\partial F}{\partial x^i}(x)\frac{d}{d\varepsilon}\left(x^i + \varepsilon\psi^i\right)\Big|_{\varepsilon=0}$$

$$= \sum_i \frac{\partial F}{\partial x^i}(x)\,\psi^i$$

if $\quad \dfrac{d}{d\varepsilon}\Big|_{\varepsilon=0} F[f + \varepsilon\psi] = \displaystyle\int d^3x\; \underbrace{(\cdots)}_{\text{call it } \frac{\delta F}{\delta f}}\;\psi(x)$

test function

$\delta F(\psi)$

$$(v + \varepsilon u)^2$$

$$= v^2 + 2\varepsilon v u + \varepsilon^2 u^2$$

$$L[q, v] = \frac{1}{2}\int_{\mathbb{R}^3} d^3x\left(v(x)^2 - \nabla q(x)\cdot\nabla q(x)\right)$$

$$\frac{\delta L}{\delta v} = ?$$

$$L[q, v+\varepsilon\psi] = \frac{1}{2}\int_{\mathbb{R}^3} d^3x \left((v(x)+\varepsilon\psi(x))^2 - \nabla q(x)\cdot\nabla q(x)\right)$$

$$= \frac{1}{2}\int_{\mathbb{R}^3} d^3x \left(v(x)^2 + 2\varepsilon v(x)\psi(x) + \varepsilon^2\psi(x)^2 - \nabla q(x)\cdot\nabla q(x)\right)$$

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} L[q, v+\varepsilon\psi] = \frac{1}{2}\int d^3x \; 2 v(x)\psi(x) \quad \rightarrow \quad \boxed{\frac{\delta L}{\delta v} = v}$$

$$\| \qquad\qquad \text{Legendre transform}$$

$$\int d^3x \; \frac{\delta L}{\delta v}(x)\,\psi(x) \qquad\qquad \boxed{p = v}$$

$$\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} L[q+\varepsilon\psi, v] = \frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\frac{1}{2}\int d^3x \left(v^2 - \nabla(q+\varepsilon\psi)\cdot\nabla(q+\varepsilon\psi)\right)$$

$$= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\frac{1}{2}\int d^3x \left(v^2 - \nabla q\cdot\nabla q - 2\varepsilon\nabla\psi\cdot\nabla q - \cancel{\varepsilon^2\nabla\psi\cdot\nabla\psi}\right)$$

$$= -\int d^3x \; \nabla\psi\cdot\nabla q \qquad \swarrow \text{ integration by parts}$$

$$= -\left(\text{boundary term}\right) + \int d^3x \; \psi\,\nabla^2 q$$

$\searrow$ 0 since ψ test function

$$\sum_i \partial_i\psi \; \partial_i q$$

$$= \sum_i \partial_i(\psi\cdot\partial_i q) - \sum_i \psi\,\partial_i^2 q$$

$$\boxed{\frac{\delta L}{\partial q} = \nabla^2 q}$$

$$\int dx \; \partial_x f = f\Big|_{-\infty}^{\infty}$$

<u>EL</u>: $$0 = \frac{\delta L}{\delta q}(\phi(t), \dot\phi(t)) - \frac{d}{dt}\frac{\delta L}{\delta v}(\phi(t), \dot\phi(t))$$

$$= \nabla^2\phi - \frac{d}{dt}(\dot\phi) = \nabla^2\phi - \ddot\phi = 0$$

<u>Legendre transform</u> define $p = \frac{\delta L}{\partial v}$ → use this to find eq. for $v = v(p)$

$v = p$

$$\boxed{H(q,p) = \int d^3x \; p\,v(p) - L(q, v(p))} \quad \text{definition}$$

$$= \int d^3x \left[p^2 - \frac{1}{2}\left(v^2 - \nabla q\cdot\nabla q\right)\right]$$

$$= \frac{1}{2}\int d^3x \left(p^2 + \nabla q\cdot\nabla q\right)$$

$$\dot q = \frac{\delta H}{\delta p}$$

$$\dot p = -\frac{\delta H}{\delta q}$$

$\mathcal{L}(q, v)$ Lagrangian $\quad q, v \in \mathbb{R}^n$

$$p_i = \frac{\partial\mathcal{L}}{\partial v^i}(q, v) \quad \left(\frac{\partial\mathcal{L}}{\partial v_i \partial v_j}\right)$$

invert to write $v^i = v^i(q,p)$

$$H = \sum_i p_i v^i - \mathcal{L}$$

Canonical quantization: $\hat{q}(\vec{x})$, $\hat{p}(\vec{x})$ operators for each $\vec{x}$

$$[\hat{q}(\vec{x}), \hat{p}(\vec{x}')] = i\,\delta(\vec{x}-\vec{x}')$$

$$[\hat{q}(\vec{x}), \hat{q}(\vec{x}')] = [\hat{p}(\vec{x}), \hat{p}(\vec{x}')] = 0$$

<u>Schrodinger:</u> $-i\frac{d}{dt}|\psi\rangle = \hat{H}|\psi\rangle \quad \rightarrow \quad U(t) = e^{-i\hat{H}t}$ $\qquad |\psi(t)\rangle = U(t)|\psi(0)\rangle$

<u>Heisenberg picture</u>: O_S Schrodinger picture op.

$$O_H(t) = U(t)^\dagger O_S U(t)$$

call
$$\begin{cases} \hat{\phi}(t,\vec{x}) := U^\dagger(t)\,\hat{q}(\vec{x})\,U(t) \\ \hat{\pi}(t,\vec{x}) := U^\dagger(t)\,\hat{p}(\vec{x})\,U(t) \end{cases}$$

$\langle \Box \hat{\phi} \rangle = 0$

<u>next step:</u> creation + annihilation

<u>Detour</u> Quantum Harmonic oscillator $\quad \hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}k^2 q^2$

$[\hat{q}_i, \hat{p}_j] = i\,\delta_{ij}$ $\qquad \omega = \sqrt{k/m}$

<u>ladder operator</u>
$$\hat{a}_i = \sqrt{\frac{m\omega}{2}}\left(\hat{q}_i + \frac{i}{m\omega}\hat{p}_i\right)$$

$$\hat{a}_i^\dagger = \sqrt{\frac{m\omega}{2}}\left(\hat{q}_i - \frac{i}{m\omega}\hat{p}_i\right)$$

$$\hat{q}_i = \sqrt{\frac{1}{2m\omega}}\left(\hat{a} + \hat{a}^\dagger\right)$$

$$\hat{p}_i = i\sqrt{\frac{2m\omega}{2}}\left(\hat{a}^\dagger - \hat{a}\right)$$

$$\rightarrow \quad \hat{H} = \omega\left(\sum_i a_i^\dagger a_i + \frac{1}{2}\right)$$

<u>eigenvectors</u> $|n_1, n_2, n_3\rangle \qquad n_i \in \mathbb{N}_0$

$a_i|0,0,0\rangle = 0 \quad \forall i \qquad$ ground state

$[a_i, a_j^\dagger] = \delta_{ij}$ $\qquad [a_i, a_j] = 0 \qquad [a_i^\dagger, a_j^\dagger] = 0$

$[H, a_i] = -\omega a_i \qquad [H, a_i^\dagger] = \omega a_i^\dagger$

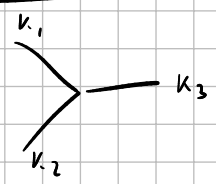
$a_1^\dagger |n_1\ n_2\ n_3\rangle \propto |n_1+1\ \ n_2\ \ n_3\rangle$

<u>plan:</u> Fourier transform

$$q(\vec{x}) = \int \frac{d^3\vec{k}}{(2\pi)^{3/2}}\, s(\vec{k})\, e^{i\vec{k}\cdot\vec{x}}$$

$$p(\vec{x}) = \int \frac{d^3\vec{k}}{(2\pi)^{3/2}}\, r(\vec{k})\, e^{i\vec{k}\cdot\vec{x}}$$

$$a(\vec{k}) = \alpha\, s(\vec{k}) + i\beta\, r(\vec{k})$$

$$[a(\vec{k}), a^\dagger(\vec{k}')] = \delta(k-k')$$

$$H = \int d^3k\, (\ldots)\, a^\dagger(k)\, a(k)$$